

Partitioning

1. What is Partitioning?

Partitioning means dividing incoming events into multiple groups based on event content, then sending each group to different hosts for processing.

A **Layer 7 Load Balancer** is used because it can inspect request content and route traffic intelligently.

2. How It Works

The load balancer checks incoming values and routes:

None

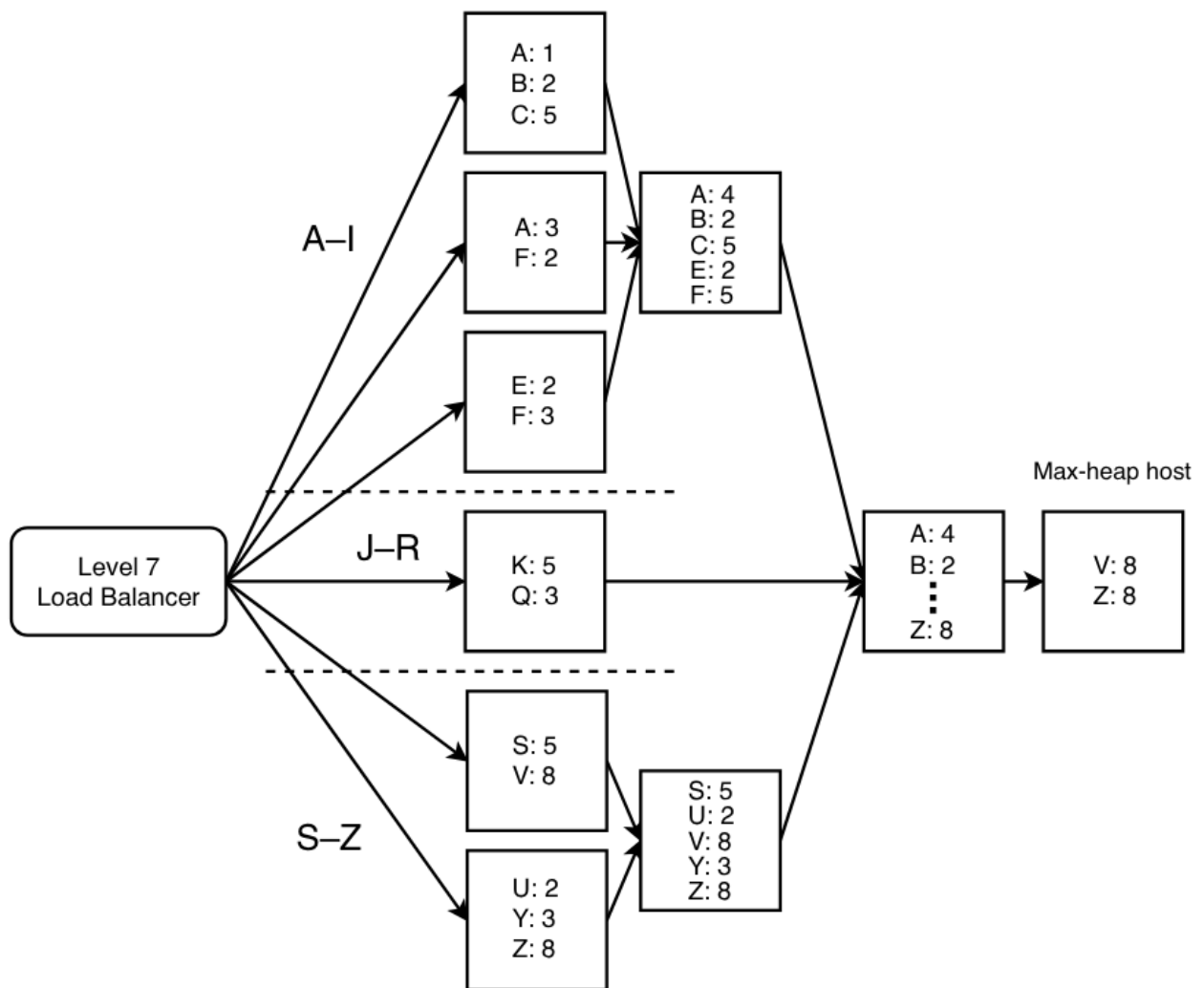
A-I → Partition 1 Hosts

J-R → Partition 2 Hosts

S-Z → Partition 3 Hosts

Each partition has its own aggregation hosts.

3. Example



Tier 1 hosts store:

None

Host A-I:

A=1

B=2

C=5

None

Host A-I:

A=3

F=2

None

Host A-I:

E=2

F=3

None

Host J-R:

K=5

Q=3

None

Host S-Z:

S=5

V=8

None

Host S-Z:

U=2

Y=3

Z=8

4. Second Layer Aggregation

Partitions combine local totals:

None

A-I:

A=4

B=2

C=5

E=2

F=5

None

J-R :

K=5

Q=3

None

S-Z :

S=5

U=2

V=8

Y=3

Z=8

5. Final Layer

All partition summaries combine into one final hash table:

None

A=4

B=2

C=5

K=5

V=8

Z=8

...

Then sent to **Max-Heap Host** to find top values.

Example:

None

Top Values:

V=8

Z=8

K=5

C=5

6. Why Use Partitioning?

Without partitioning:

None

All events go to same hosts

Hotspots occur

Overload risk

With partitioning:

None

Traffic split across partitions

Parallel processing

Better scalability

7. Uneven Traffic Problem

Traffic usually follows non-uniform distribution.

Some partitions become hot.

Example:

None

A-I receives more traffic

J-R less traffic

S-Z medium traffic

So allocate hosts differently:

None

A-I = 3 hosts

J-R = 1 host

S-Z = 2 hosts

8. Dynamic Repartitioning

Can redesign partitions:

Old:

None

A-I

J-R

S-Z

New:

None

A-B, D-F

C, G-J

K-S

T-Z

This balances load better.

9. Advantages

None

Parallel processing

Removes hot nodes

Scales horizontally

Flexible host allocation

Fast top-k processing

10. Disadvantages

None

More routing complexity

Need smart load balancer

Repartitioning may be required

Skewed traffic still possible

11. Best Use Cases

- Search terms counting
- Top trending hashtags
- Clickstream categories
- Product popularity ranking
- Sensor event grouping

12. Interview Talking Point

If one node gets overloaded:

None

Use content-based partitioning

Route A-I / J-R / S-Z separately

Allocate more hosts to hot partitions

Strong HLD answer.

13. One-Line Summary

Partitioning uses a Layer 7 load balancer to divide events into groups based on content and process them on separate host pools, improving scalability and preventing hotspots.